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## 1 Reactions and pathways

This document is the registry of (photo)electrocatalytic reactions and reaction pathways implemented in `codes/files/adsorbates.py`, plus a list of natural extensions that fit the same template. All implemented reactions follow the computational hydrogen electrode (CHE) framework (Nørskov et al. 2004).

The dispatch entry point is `generate_adsorbed_structures(reaction, pathway_name)` in `codes/files/adsorbates.py`.

## 1.1 Molecular reference structures

Gas-phase / solid-state reference cells live in `codes/files/molecular_references/` as VASP POSCARs and are loaded once at import time. Provenance:

| Ref   | Source                                 | Notes                                 |
|-------|----------------------------------------|---------------------------------------|
| h2    | Materials Project mp-* (P2_12_12_1)    | Z=4                                   |
| h2o   | Materials Project mp-* (Cmc2_1)        | Z=8 (ice-like)                        |
| n2    | Materials Project mp-* (P2_13)         | Z=4 (alpha-N2)                        |
| nh3   | Materials Project mp-* (P2_13)         | Z=4 (cubic ammonia)                   |
| n2o   | Materials Project mp-* (Pnma)          | Z=4                                   |
| co2   | Materials Project mp-* (Pa-3)          | Z=4 (dry ice)                         |
| ch4   | Materials Project mp-* (I-43m)         | Z=2                                   |
| ch3oh | hand-built gas-phase, 15 Å vacuum cube | equilibrium geometry, single molecule |
| cl2   | Materials Project mp-22848 (Cmce)      | Z=2, stable                           |
| o2    | Materials Project mp-723285 (P4_12_12) | Z=8, near-stable                      |
| h2o2  | Materials Project mp-28015 (P4_12_12)  | Z=16                                  |

When a new gas-phase reference is needed, prefer the Materials Project crystal (matches the project's per-formula-unit-energy convention) and use a vacuum-cube single molecule only when no MP entry is suitable.

## 2 Currently implemented reactions

### 2.1 OER — oxygen evolution

Dispatch key: `{"OER": None}` (pathway name not used).

Triggers OER computation and stores the overpotential in PostgreSQL. Single  $4e^-$  associative pathway:  
 $* \rightarrow *OH \rightarrow *O \rightarrow *OOH \rightarrow *$  releasing  $O_2$ .

### 2.2 References

Nørskov et al. *J. Phys. Chem. B* **108**, 17886 (2004). Man et al. *ChemCatChem* **3**, 1159 (2011) — universal scaling on perovskites. Rossmeisl et al. *J. Electroanal. Chem.* **607**, 83 (2007).

### 2.3 CO2RR — CO<sub>2</sub> electroreduction

Dispatch key: `{"CO2RR": "<pathway_name>"}`.

### 2.4 Pathways implemented

- 'co2\_to\_co':  $CO_2 \rightarrow *COOH \rightarrow *CO \rightarrow CO(g)$  (Au, Ag, Zn)
- 'co2\_to\_hcooh':  $CO_2 \rightarrow *OCHO \rightarrow HCOOH(aq)$  (formate; Pd, In, Sn)
- 'co\_to\_ch4':  $*CO \rightarrow *CHO \rightarrow *CHOH \rightarrow *CH \rightarrow *CH_2 \rightarrow *CH_3 \rightarrow CH_4(g)$  (Cu)
- 'co\_to\_ch3oh':  $*CO \rightarrow *CHO \rightarrow *CHOH \rightarrow *CH_2OH \rightarrow CH_3OH(g)$  (Cu, Mo)
- 'co2\_to\_ch4': full  $CO_2 \rightarrow CH_4$  chain on Cu
- 'co2\_to\_ch3oh': full  $CO_2 \rightarrow CH_3OH$  chain on Cu/Mo
- 'co2\_to\_c2h4':  $2 *CO \rightarrow *OCCO \rightarrow \dots \rightarrow C_2H_4(g)$  (Cu, C-C coupling)

## 2.5 References

Peterson et al. *Energy Environ. Sci.* **3**, 1311 (2010). Kuhl et al. *J. Am. Chem. Soc.* **136**, 14107 (2014). Montoya et al. *ChemSusChem* **8**, 2180 (2015). Goodpaster et al. *J. Phys. Chem. Lett.* **7**, 1471 (2016).

## 2.6 NO<sub>x</sub>RR — NO<sub>x</sub> electroreduction

Dispatch key: {"NO<sub>x</sub>RR": "<pathway\_name>"}. NO<sub>x</sub> covered: NO, NO<sub>2</sub>, NO<sub>3</sub><sup>-</sup>.

## 2.7 Pathways implemented

- 'no\_dissociative': \*NO → \*N + \*O → N<sub>2</sub>(g) (Ru, Rh, Ir)
- 'no\_to\_nh3\_noh': \*NO → \*NOH → \*N → \*NH<sub>2</sub> → NH<sub>3</sub> (Cu, Fe)
- 'no\_to\_nh3\_nhoh': \*NO → \*NOH → \*NHOH → \*NH<sub>2</sub>OH → NH<sub>3</sub> (Cu, hydroxylamine route)
- 'no\_to\_n2o': 2 \*NO → \*ONNO → N<sub>2</sub>O + \*O (Pt, Pd automotive)
- 'no2\_to\_no': \*NO<sub>2</sub> → \*NO + \*O (prereduction step)
- 'no3\_to\_nh3': \*NO<sub>3</sub> → \*NO<sub>2</sub> → \*NO → ... → NH<sub>3</sub> (eNO<sub>3</sub>RR, Cu)
- 'no3\_to\_n2': \*NO<sub>3</sub> → \*NO<sub>2</sub> → \*NO → \*N → N<sub>2</sub> (eNO<sub>3</sub>RR, Ru)

## 2.8 References

Gao et al. *Nat. Chem.* **9**, 547 (2017). Liu et al. *Nat. Commun.* **12**, 5797 (2021). Wang et al. *J. Am. Chem. Soc.* **142**, 5702 (2020). van 't Veer et al. *J. Phys. Chem. C* **124**, 22 (2020). Pérez-Ramírez & López *Nat. Catal.* **2**, 971 (2019).

## 2.9 CER — chlorine evolution

Dispatch key: {"CER": "<pathway\_name>"}. CHE referenced to the Cl<sup>-</sup>/Cl<sub>2</sub> couple (E° = 1.36 V vs SHE in 1 M HCl). Electrochemical steps consume Cl<sup>-</sup> instead of H<sup>+</sup> (protons=0, electrons=-1).

## 2.10 Pathways implemented

- 'volmer\_tafel': \* + Cl<sup>-</sup> → \*Cl, then 2 \*Cl → Cl<sub>2</sub>(g) + 2 \* (Pt, Ru, Pd)
- 'volmer\_heyrovsky': \* + Cl<sup>-</sup> → \*Cl, then \*Cl + Cl<sup>-</sup> → Cl<sub>2</sub>(g) + \* (RuO<sub>2</sub>, IrO<sub>2</sub>)
- 'krishtalik': \*O + Cl<sup>-</sup> → \*OCl, then \*OCl + Cl<sup>-</sup> → Cl<sub>2</sub> + \*O (RuO<sub>2</sub>, IrO<sub>2</sub>, Co<sub>3</sub>O<sub>4</sub>)

## 2.11 References

Hansen et al. *Phys. Chem. Chem. Phys.* **12**, 283 (2010). Exner et al. *Electrocatalysis* **6**, 163 (2015). Exner et al. *ChemElectroChem* **3**, 1607 (2016). Karlsson & Cornell *Chem. Rev.* **116**, 2982 (2016). Lim et al. *Nat. Commun.* **11**, 412 (2020). Kim et al. *Angew. Chem. Int. Ed.* **64**, e202417293 (2025).

## 2.12 HER — hydrogen evolution

Dispatch key: {"HER": "<pathway\_name>"}. Foundational reaction; \*H is the single intermediate, |ΔG<sub>H</sub>\*| ≈ 0 marks the volcano apex.

## 2.13 Pathways implemented

- 'volmer\_tafel': \* + H<sup>+</sup> + e<sup>-</sup> → \*H, then 2 \*H → H<sub>2</sub>(g) + 2 \* (Pt, Pd, Ir)
- 'volmer\_heyrovsky': \* + H<sup>+</sup> + e<sup>-</sup> → \*H, then \*H + H<sup>+</sup> + e<sup>-</sup> → H<sub>2</sub>(g) + \* (MoS<sub>2</sub>, Au, Ag, Ni-Mo)

## 2.14 References

Nørskov et al. *J. Electrochem. Soc.* **152**, J23 (2005). Greeley et al. *Nat. Mater.* **5**, 909 (2006). Conway & Tilak *Electrochim. Acta* **47**, 3571 (2002). Skúlason et al. *J. Phys. Chem. C* **114**, 18182 (2010).

## 2.15 ORR — oxygen reduction

Dispatch key: {"ORR": "<pathway\_name>"}. Cathodic counterpart of OER. Same \*OH-\*OOH scaling relation (~3.2 eV) caps activity at  $\eta \approx 0.3$ –0.4 V.

## 2.16 Pathways implemented

- '4e\_associative' :  $O_2 \rightarrow *O_2 \rightarrow *OOH \rightarrow *O \rightarrow *OH \rightarrow H_2O$  (Pt, Pd, Pt<sub>3</sub>Ni)
- '4e\_dissociative' :  $O_2 \rightarrow 2 *O \rightarrow *OH \rightarrow H_2O$  (Pt(111) at low T)
- '2e\_to\_h2o2' :  $O_2 \rightarrow *O_2 \rightarrow *OOH \rightarrow H_2O_2(aq)$  (Au, Hg, Co-N<sub>4</sub>, Fe-N<sub>4</sub>)

## 2.17 References

Nørskov et al. *J. Phys. Chem. B* **108**, 17886 (2004). Stamenkovic et al. *Nat. Mater.* **6**, 241 (2007). Greeley et al. *Nat. Chem.* **1**, 552 (2009). Siahrostami et al. *Nat. Mater.* **12**, 1137 (2013). Kulkarni et al. *Chem. Rev.* **118**, 2302 (2018).

## 2.18 NRR — nitrogen reduction

Dispatch key: {"NRR": "<pathway\_name>"}. Electrochemical analog of Haber-Bosch. Strongly limited by HER competition and by weak \*N<sub>2</sub> binding on most metals.

## 2.19 Pathways implemented

- 'distal' :  $*N_2 \rightarrow *NNH \rightarrow *NNH_2 \rightarrow *N + NH_3 \rightarrow *NH \rightarrow *NH_2 \rightarrow NH_3$  (Ru, Mo, Re, Mo-N<sub>3</sub>)
- 'alternating' :  $*N_2 \rightarrow *NNH \rightarrow *NHNH \rightarrow *NHNH_2 \rightarrow *N_2H_4 \rightarrow *NH_2 + NH_3 \rightarrow *NH_3 \rightarrow NH_3$  (Fe, Co, Fe-N<sub>4</sub>)
- 'dissociative' :  $N_2 \rightarrow 2 *N \rightarrow 2 *NH \rightarrow 2 *NH_2 \rightarrow 2 NH_3$  (Fe, Ru, Os; needs strong \*N binding)

## 2.20 References

Skúlason et al. *Phys. Chem. Chem. Phys.* **14**, 1235 (2012). Montoya et al. *ChemSusChem* **8**, 2180 (2015). Singh et al. *ACS Catal.* **7**, 706 (2017). Andersen et al. *Nature* **570**, 504 (2019). Suryanto et al. *Nat. Catal.* **2**, 290 (2019).

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## 3 Planned / candidate reactions

The following all fit the same ReactionPathway / \_create\_adsorbate\_with\_dummy template and would be drop-in additions. Effort estimates are relative to the HER block (smallest existing reaction; ~100 lines).

### 3.1 HOR — hydrogen oxidation

Reverse of HER (Volmer + Heyrovsky/Tafel run backwards). Effort: tiny; just reversed pathways, no new geometry. Useful for fuel-cell anode work.

### 3.2 References

Sheng et al. *J. Electrochem. Soc.* **157**, B1529 (2010). Strmcnik et al. *Nat. Chem.* **5**, 300 (2013). Durst et al. *Energy Environ. Sci.* **7**, 2255 (2014).

### 3.3 AOR — ammonia oxidation

$2 \text{NH}_3 + 6 \text{OH}^- \rightarrow \text{N}_2 + 6 \text{H}_2\text{O} + 6 \text{e}^-$ . Dehydrogenation cascade  $^*\text{NH}_3 \rightarrow ^*\text{NH}_2 \rightarrow ^*\text{NH} \rightarrow ^*\text{N} \rightarrow \text{N}_2$ ; reuses every intermediate already in NOxRR. Effort: tiny. Important for direct-ammonia fuel cells.

### 3.4 References

Vidal-Iglesias et al. *J. Catal.* **233**, 237 (2005). Bunce & Bejan *Electrochim. Acta* **56**, 8085 (2011). Boggs & Botte *J. Power Sources* **192**, 573 (2009). Rosca et al. *Chem. Rev.* **109**, 2209 (2009).

### 3.5 MOR — methanol oxidation

$\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 6 \text{H}^+ + 6 \text{e}^-$ . Dehydrogenation through  $^*\text{CH}_3\text{OH} \rightarrow ^*\text{CH}_2\text{OH} \rightarrow ^*\text{CHOH} \rightarrow ^*\text{CO} \rightarrow \text{CO}_2$ , with the well-known  $^*\text{CO}$  poisoning detour. Reuses CO, CHO, CHOH,  $\text{CH}_2\text{O}$ ,  $\text{CH}_2\text{OH}$  from CO2RR. Effort: small. New ref: gas-phase  $\text{CH}_3\text{OH}$  already present.

### 3.6 References

Hamnett *Catal. Today* **38**, 445 (1997). Yu et al. *J. Phys. Chem. C* **116**, 10906 (2012). Cao et al. *Chem. Soc. Rev.* **35**, 1230 (2006). Wasmus & Küver *J. Electroanal. Chem.* **461**, 14 (1999).

### 3.7 FAOR — formic acid oxidation

$\text{HCOOH} \rightarrow \text{CO}_2 + 2 \text{H}^+ + 2 \text{e}^-$ . Direct ( $^*\text{HCOO} \rightarrow \text{CO}_2$ ) vs indirect ( $^*\text{COOH} \rightarrow ^*\text{CO}$  poison  $\rightarrow \text{CO}_2$ ) bifurcation. Reuses OCHO/COOH/CO from CO2RR. Effort: small. New ref: HCOOH crystal (look up MP).

### 3.8 References

Capon & Parsons *J. Electroanal. Chem.* **44**, 1 (1973) — original dual-path. Cuesta et al. *Phys. Chem. Chem. Phys.* **13**, 20091 (2011). Yu & Pickup *J. Power Sources* **182**, 124 (2008). Chen et al. *Angew. Chem. Int. Ed.* **45**, 981 (2006).

### 3.9 BER — bromine evolution

$2 \text{Br}^- \rightarrow \text{Br}_2 + 2 \text{e}^-$ . Direct analog of CER with Br substituted for Cl. Volmer-Tafel, Volmer-Heyrovsky, and Krishtalik ( $^*\text{OBr}$ ) pathways. Effort: small (copy of CER). New refs:  $^*\text{Br}$ ,  $^*\text{OBr}$  adsorbates and  $\text{Br}_2$  crystal.

### 3.10 References

Vos & Koper *J. Electroanal. Chem.* **819**, 260 (2018). Karlsson & Cornell *Chem. Rev.* **116**, 2982 (2016). Tomashov & Strukov *Russ. J. Phys. Chem.* **35**, 1112 (1961) — original kinetics.

### 3.11 UOR — urea oxidation

$\text{CO}(\text{NH}_2)_2 + 6 \text{OH}^- \rightarrow \text{N}_2 + \text{CO}_2 + 5 \text{H}_2\text{O} + 6 \text{e}^-$ . Direct vs Ni(III)-mediated (indirect) mechanism. Effort: medium-large. Adds urea-specific intermediate geometries ( $^*\text{urea}$ ,  $^*\text{NH}_2\text{CO}$ ,  $^*\text{NHCO}$ , ...) and a urea gas-phase reference. Practical relevance: wastewater +  $\text{H}_2$  co-generation.

### 3.12 References

Boggs et al. *Chem. Commun.* 4859 (2009). Wang et al. *J. Mater. Chem. A* **5**, 3208 (2017). Yan et al. *Appl. Catal. B* **122-123**, 73 (2012). Vedharathinam & Botte *Electrochim. Acta* **108**, 660 (2013).

### 3.13 EOR — ethanol oxidation

$C_2H_5OH \rightarrow CO_2$  (full oxidation) or  $CH_3COOH$  (partial).  $C_1$  vs  $C_2$  branching with C-C bond cleavage as the central question. Effort: large; the mechanism is genuinely unsettled and the intermediate library is extensive (\* $CH_3CH_2OH$ , \* $CH_3CHO$ , \* $CH_3COO$ , \* $CHCO$ , ...).

### 3.14 References

Lai et al. *ACS Catal.* **2**, 1042 (2012). Bayer et al. *Appl. Catal. B* **138-139**, 313 (2013). Ferrin & Mavrikakis *J. Am. Chem. Soc.* **131**, 14381 (2009). Wang et al. *J. Phys. Chem. C* **116**, 6675 (2012).

### 3.15 Photo-driven variants

Photocatalytic water splitting (HER + OER on a semiconductor),  $CO_2$  photoreduction,  $N_2$  photofixation. The pathway machinery is identical to the electrochemical counterparts; the photo aspect enters as a thermodynamic gate (semiconductor band edges vs intermediate energies) and belongs in the analysis layer downstream of `adsorbates.py`. Likely no new code in `adsorbates.py` — just a flag downstream.

### 3.16 References

Walter et al. *Chem. Rev.* **110**, 6446 (2010). Fujishima & Honda *Nature* **238**, 37 (1972). White et al. *Chem. Rev.* **115**, 12888 (2015) —  $CO_2$  photoreduction. Comer et al. *Joule* **3**, 1578 (2019) —  $N_2$  photofixation.

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## 4 Overpotential calculators

One module per reaction under `workchains/`. Each implements the CHE model with the same return shape:

```
overpotential, dg_steps, dg_cumulative = calculate_<rxn>_overpotential(
    adsorption_energies, pathway_name, method, func)
```

where `adsorption_energies` is a dict keyed by \*X adsorbate names, `pathway_name` matches the pathway keys in this document, and (`method`, `func`) pick the gas-phase reference set from `workchains/references.yaml` (e.g. ("`dft`", "`r2SCAN`"), ("`uPET`", "`r2SCAN`").

| Reaction | Module                           | Pathway keys                                                                                                                                                                                    |
|----------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OER      | <code>workchains/oer.py</code>   | (4e- mechanism, fixed)                                                                                                                                                                          |
| CO2RR    | <code>workchains/co2rr.py</code> | <code>co2_to_co</code> , <code>co2_to_hcooh</code> , <code>co2_to_ch4</code> , <code>co2_to_ch3oh</code> ,<br><code>co2_to_ch4</code> , <code>co2_to_ch3oh</code> , <code>co2_to_c2h4</code>    |
| NOXRR    | <code>workchains/noxrr.py</code> | <code>no_dissociative</code> , <code>no_to_nh3_noh</code> , <code>no_to_nh3_nhoh</code> ,<br><code>no_to_n2o</code> , <code>no2_to_no</code> , <code>no3_to_nh3</code> , <code>no3_to_n2</code> |
| CER      | <code>workchains/cer.py</code>   | <code>volmer_tafel</code> , <code>volmer_heyrovsky</code> , <code>krishtalik</code>                                                                                                             |
| HER      | <code>workchains/her.py</code>   | <code>volmer_tafel</code> , <code>volmer_heyrovsky</code>                                                                                                                                       |
| ORR      | <code>workchains/orr.py</code>   | <code>4e_associative</code> , <code>4e_dissociative</code> , <code>2e_to_h2o2</code>                                                                                                            |
| NRR      | <code>workchains/nrr.py</code>   | <code>distal</code> , <code>alternating</code> , <code>dissociative</code>                                                                                                                      |

**ZPE corrections** are loaded once at module import via `load_zpe(reaction)` from `workchains/zpe_correction`. That file covers every adsorbate and gas-phase species touched by these pathways; see the file header for the convention (full harmonic ZPE per species, not a delta against ZPE-folded references).

**Gas-phase references** are pulled per-call via `load_references(method, func)` from `workchains/references`. Prerequisites flagged by the new modules:

- `cer.py` requires a `Cl2:` entry in every `(method, func)` block.
- `orr.py`'s `2e_to_h2o2` pathway requires an `H2O2:` entry.

The CER `equilibrium_potential = 1.36` is vs SHE (the  $\text{Cl}^-/\text{Cl}_2$  couple), not vs RHE — downstream code reading this value should account for the reference difference.

The pathway dicts in the per-reaction modules use the same stoichiometric convention as `co2rr.py` (positive = product, negative = reactant; each electrochemical step adds  $\text{H}_2$ :  $+1/2$  for proton-coupled chemistry or  $\text{Cl}_2$ :  $+1/2$  for CER's chloride-coupled chemistry).

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## 5 Nano particles

Dispatch key: `'nano_particles': False / True`.

The key has to be present or the frontend update will fail. With `True`, the catalysis `WorkChain` is bypassed entirely and nano-particle generation runs instead. More documentation will follow.